

EXP 8

Program

Server program

```
import java.io.*;

import java.net.*;

public class ConcurrentFileServer {

    private static final int PORT = 5000;

    public static void main(String[] args) {

        try (ServerSocket serverSocket = new ServerSocket(PORT)) {

            long pid = ProcessHandle.current().pid();

            System.out.println("Concurrent File Server Started...");

            System.out.println("Server PID: " + pid);

            System.out.println("Listening on port " + PORT);

            while (true) {

                Socket clientSocket = serverSocket.accept();

                System.out.println("New client connected: " + clientSocket.getInetAddress());

                // Create new thread for each client - This makes it concurrent

                new Thread(new ClientHandler(clientSocket, pid)).start();
            }
        }
    }
}
```

```
    }  
  
    } catch (IOException e) {  
  
        e.printStackTrace();  
  
    }  
  
}  
  
}
```

```
class ClientHandler implements Runnable {  
  
    private Socket clientSocket;  
  
    private long serverPid;  
  
  
    public ClientHandler(Socket socket, long pid) {  
  
        this.clientSocket = socket;  
  
        this.serverPid = pid;  
  
    }  
  

```

```
    @Override  
  
    public void run() {  
  
        try (  
  
            BufferedReader in = new BufferedReader(new  
InputStreamReader(clientSocket.getInputStream()));
```

```

        PrintWriter out = new PrintWriter(clientSocket.getOutputStream(), true)
    ){

        // 1. Read filename from client

        String fileName = in.readLine();

        System.out.println "[" + Thread.currentThread().getId() + "] Requested file: " +
        fileName);

        File file = new File(fileName);

        // 2. Send PID first

        out.println("Server PID: " + serverPid);

        out.println("Servicing Thread ID: " + Thread.currentThread().getId());

        if (file.exists() && file.isFile()) {

            out.println("--- FILE FOUND ---");

            BufferedReader fileReader = new BufferedReader(new FileReader(file));

            String line;

            while ((line = fileReader.readLine()) != null) {

                out.println(line);

            }

            fileReader.close();

```

```

    } else {

        out.println("--- ERROR: File " + fileName + " does not exist on server. ---");

    }

    out.println("EOF"); // End of file marker


    } catch (IOException e) {

        e.printStackTrace();

    } finally {

        try { clientSocket.close(); } catch (IOException e) {}

    }

}

```

Client program

```

import java.io.*;

import java.net.*;

import java.util.Scanner;


public class FileClient {

    public static void main(String[] args) {

```

```
String serverAddress = "localhost";

int port = 5000;

try (Socket socket = new Socket(serverAddress, port);

    BufferedReader in = new BufferedReader(new
InputStreamReader(socket.getInputStream()));

    PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

    Scanner sc = new Scanner(System.in)) {

    System.out.print("Enter filename to request from server: ");

    String fileName = sc.nextLine();

    // Send filename to server

    out.println(fileName);

    // Receive and display response

    String responseLine;

    System.out.println("\n--- Response from Server ---");

    while ((responseLine = in.readLine()) != null) {

        if (responseLine.equals("EOF")) break;

        System.out.println(responseLine);
```

```
}
```

```
} catch (IOException e) {
```

```
    System.out.println("Could not connect to server.");
```

```
    e.printStackTrace();
```

```
}
```

```
}
```

```
}
```

```
sjcet@sjcet-H81M-H-M-2:~$ mkdir exp8
sjcet@sjcet-H81M-H-M-2:~$ cd exp8
```

```
sjcet@sjcet-H81M-H-M-2:~/exp8$ gedit ConcurrentFileServer.java
sjcet@sjcet-H81M-H-M-2:~/exp8$ javac ConcurrentFileServer.java
sjcet@sjcet-H81M-H-M-2:~/exp8$ gedit FileClient.java
sjcet@sjcet-H81M-H-M-2:~/exp8$ javac FileClient.java
sjcet@sjcet-H81M-H-M-2:~/exp8$ java ConcurrentFileServer
Concurrent File Server Started...
Server PID: 15164
Listening on port 5000
New client connected: /127.0.0.1
[13] Requested file: test.txt
New client connected: /127.0.0.1
[14] Requested file: text.txt
```

```
sjcet@sjcet-H81M-H-M-2:~/exp8$ java FileClient
Enter filename to request from server: test.txt
```

--- Response from Server ---

Server PID: 15164

Servicing Thread ID: 13

--- FILE FOUND ---

Hello from the server!

This is my test file.

```
sjcet@sjcet-H81M-H-M-2:~/exp8$ java FileClient
Enter filename to request from server: text.txt
```

--- Response from Server ---

Server PID: 15164

Servicing Thread ID: 14

--- ERROR: File 'text.txt' does not exist on server. ---

```
sjcet@sjcet-H81M-H-M-2:~/exp8$
```